

Taylor polynomial

$$f: I \rightarrow \mathbb{R}$$

$$x_0 \in I$$

f diff. around x_0 (m times)

- we need to approximate f as well as we can -

$$\left. \begin{array}{l} P(x_0) = f(x_0) \\ P'(x_0) = f'(x_0) \\ \vdots \\ P^{(m)}(x_0) = f^{(m)}(x_0) \end{array} \right\} \Rightarrow n+1 \text{ equations} = n+1 \text{ nr. of "unknowns"}$$

$\Downarrow$
degree m .

(ex: $x^2 + 2x + 1 \rightarrow \text{degree} = 2$
 $\rightarrow \text{unknowns} = 3$)

Consider the following polynomial P of degree at most m :

$$P(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \dots + a_m(x-x_0)^m$$

- Goal: find P 's coefficients -

$$P(x_0) = f(x_0) = a_0$$

$$\Rightarrow a_0 = f(x_0)$$

$\leadsto$ INTERPOLATION

(the poly = f)

$$P'(x_0) = f'(x_0) = [a_1 + 2a_2(x-x_0) + \dots + m \cdot a_m(x-x_0)^{m-1}]_{x=x_0} = a_1$$

$$\Rightarrow a_1 = f'(x_0)$$

$$P''(x_0) = f''(x_0) = [2a_2 + \dots + m \cdot (m-1) \cdot a_m(x-x_0)^{m-2}]_{x=x_0} = 2 \cdot a_2$$

$$\Rightarrow a_2 = \frac{f''(x_0)}{2}$$

- with each interpolation we discover a new coefficient.

$\vdots$

$$P^{(m)}(x_0) = f^{(m)}(x_0) = m! \cdot a_m$$

$$\Rightarrow a_m = \frac{f^{(m)}(x_0)}{m!}$$

$\Rightarrow$ There is a unique polynomial with those coefficients:

$$(*) \quad P(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2} \cdot (x-x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!} \cdot (x-x_0)^n$$

↓

$T_n(x)$ = the Taylor polynomial of degree n centered around x_0 .

↓

this polynomial is approximating our function. How?

↓

Linear approx: „we linearize the function“.

↳ of f around x_0 is given by T_1 :

$$\boxed{f(x) \simeq f(x_0) + f'(x_0) \cdot (x-x_0)} \quad \text{TANGENT LINE}$$

Quadratic approx:

↳ of f around x_0 is given by T_2 :

$$f(x) \simeq f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2} (x-x_0)^2$$

!
 The closer x is to x_0
 and
 The higher the degree of T_m
 } $\Rightarrow$ the better $T_m(x)$ approximates $f(x)$.

① Construct the Taylor polynomial for $f(x) = e^x$, and $x_0 = 0$.

- we need those coefficients $\rightarrow$ derivatives from (*) -

$$(*) : T_m(x) = f(x_0) + \overset{!}{f'(x_0)}(x-x_0) + \frac{\overset{!}{f''(x_0)}}{2}(x-x_0)^2 + \dots + \frac{\overset{!}{f^{(m)}(x_0)}}{m!}(x-x_0)^m$$

$$= \sum_{k=0}^m \frac{\overset{!}{f^{(k)}(x_0)}}{k!} (x-x_0)^k$$

$$= \sum_{k=0}^m \frac{(x-x_0)^k}{k!}$$

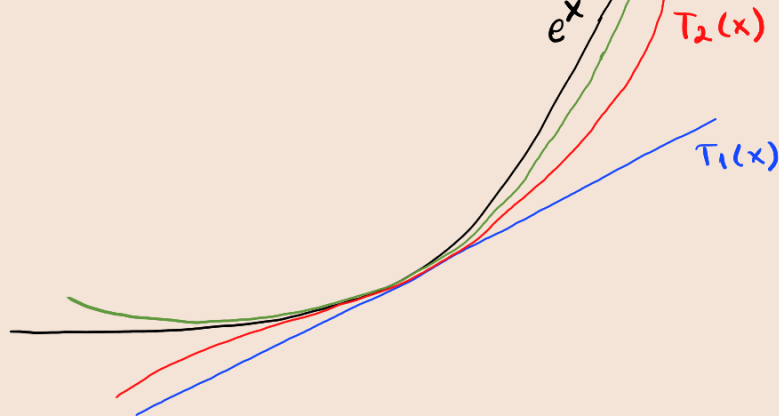
$$= 1 + x + \frac{x^2}{2!} + \dots + \frac{x^m}{m!} \quad \leadsto \text{The Taylor polynomial of } f(x) = e^x.$$

$$f'(x) = e^x$$

$$f^{(k)}(x) = e^x$$

$$f^{(k)}(\underline{x_0}) = f^{(k)}(\underline{0}) = e^0 = 1.$$

/ $T_3(x)$



- Now I would like to measure how close I am to the function (error) -

Def: $R_m(x) := f(x) - T_m(x)$ → The remainder when approximating f by T_m .

Theorem (Taylor-Lagrange): $R_m(x) = \frac{f^{(m+1)}(c)}{(m+1)!} \cdot (x-x_0)^{m+1}$ → The remainder in Lagrange's form
(proof with mean/Lagrange)
 intermediate point; $c \in (x, x_0)$

(*) $f(x) = T_m(x) + \frac{f^{(m+1)}(c)}{(m+1)!} (x-x_0)^{m+1}$ → Taylor's formula with Lagrange remainder

Def: $I \subseteq \mathbb{R}$, $f: I \rightarrow \mathbb{R}$, ∞ diff. For $x_0 \in I$ and $x \in \mathbb{R}$,

the series $\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$ → Taylor series of f around x_0

If $\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$ converges and $= f(x)$ $\Rightarrow$ f can be expanded in a Taylor series around x_0
↓
TAYLOR EXPANSION.

! A Taylor series converges to $f(x)$ ($\Leftrightarrow$) $f(x) - T_m(x) = R_m(x) \rightarrow 0$ as $m \rightarrow \infty$.

- What if we keep increasing the degree? -

$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ → the Taylor series around 0 $\xrightarrow{\text{conv.}}$ f .
↓
Maclaurin series.

$$\textcircled{2} \quad e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \dots + \frac{x^n}{n!} + \dots, \quad (\forall) x \in \mathbb{R}.$$

because $f^{(n)}(0) = (e^x)^{(n)} \big|_{x=x_0=0} = e^0 = 1.$

Why does it converge to e^x ? $\lim (*) \Rightarrow \dots \Rightarrow$ Taylor poly = Taylor series?

→ Show that for a certain function the remainder goes to 0, $n \rightarrow \infty$.

Taylor series for e^x : $\boxed{\sum_{n=0}^{\infty} \frac{x^n}{n!} \stackrel{?}{=} e^x}$

$(*)$: $e^x = T_n(x) + R_n(x)$

$$= \sum_{k=0}^n \frac{x^k}{k!} + \frac{e^c}{(n+1)!} \cdot x^{n+1}, \quad c \in (x, x_0)$$

$$\lim_n R_n(x) = \lim_n \frac{e^c \cdot x^{n+1}}{(n+1)!} = e^c \cdot \lim_n \frac{x^{n+1}}{(n+1)!} \xrightarrow{a_n \rightarrow \text{Ratio test} \rightarrow 0} \text{ct. } 0 = 0.$$

$$\lim_n (*) \Rightarrow e^x = \lim_n T_n(x) + \underbrace{\lim_n R_n(x)}_0 = \lim_n T_n(x) =$$

$$= \lim_n \sum_{k=0}^n \frac{x^k}{k!}$$

$$= \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

$$\frac{a_{n+1}}{a_n} = \frac{x^{n+2}}{(n+2)!} \cdot \frac{(n+1)!}{x^{n+1}} = \frac{x}{n+2} \rightarrow 0 < 1. \Rightarrow \lim_n a_n = 0, \quad (\forall) x \in \mathbb{R}.$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot x^{2n+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \quad \rightarrow x: = -x \Rightarrow \sin \rightarrow \text{odd}.$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \cdot x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \quad \rightarrow x: = -x \Rightarrow \cos \rightarrow \text{even}.$$

If abs. conv. $\Rightarrow$ diff. term by term $\Rightarrow (\sin x)' = \dots = \cos x$
 $(\cos x)' = \dots = -\sin x.$

TEMPLATE:

$$f(x) = T_m(x) + R_m(x)$$

$$\lim_m f(x) = \lim_m T_m(x) + \lim_m R_m(x)$$

$$T_m(x) = \sum_{k=0}^m \frac{f^{(k)}(x_0)}{k!} \cdot (x-x_0)^k$$

$$R_m(x) = \frac{f^{(m+1)}(c)}{(m+1)!} (x-x_0)^{m+1}, \quad c \in (x, x_0)$$

$$\text{If } \lim_m R_m(x) \rightarrow 0, \text{ then } \lim_m f(x) = \lim_m T_m(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$$

which is the Taylor series.

$$f(x) = \sin x \quad | \quad x_0 = 0 \quad \Rightarrow \text{the derivatives of order } m:$$

$$f(0) = \sin 0 = 0$$

$$f'(0) = \cos 0 = 1$$

$$f''(0) = -\sin 0 = 0$$

$$f^{(3)}(0) = -\cos 0 = -1$$

$$f^{(4)}(0) = \sin 0 = 0$$

⋮

$$\text{Pattern: } f^{(2k)}(0) = 0$$

$$\underline{f^{(2k+1)}(0) = (-1)^k}$$

$$\begin{aligned} T_{2m}(x) &\stackrel{?}{=} T_{2m+1}(x) = \sum_{k=0}^{2m+1} \frac{f^{(k)}(0)}{k!} (x-0)^k = \sum_{k=0}^m \frac{\overbrace{f^{(2k+1)}(0)}^{(-1)^k}}{(2k+1)!} \cdot x^{2k+1} = \\ &= \sum_{k=0}^m \frac{(-1)^k \cdot x^{2k+1}}{(2k+1)!} \quad \text{The Taylor polynomial} = T_{2m+1}(x). \end{aligned}$$

$$\lim_m R_m(x) \stackrel{?}{\rightarrow} 0$$

$$R_{2m+1}(x) = \frac{f^{(2m+2)}(c)}{(2m+2)!} \cdot x^{2m+2} = \frac{\sin^{(2m+2)}(c)}{(2m+2)!} \cdot x^{2m+2} =$$

$$= \underbrace{\sin^{(2m+2)}(c)}_{\downarrow} \cdot \frac{x^{2m+2}}{(2m+2)!}$$

$$\in [-1, 1]$$

Ratio $\rightarrow 0$

= bounded. 0

= 0

$$\Rightarrow \lim_n R_{2n+1}(x) = 0$$

$$\Rightarrow \lim_n T_{2n+1}(x) = f(x) = \sin x$$

$$\sum_{k=0}^{\infty} \frac{(-1)^k \cdot x^{2k+1}}{(2k+1)!} = \sin x$$

! The function = the limit of the Taylor series $\rightarrow$ almost for (*) f .

An example of a function that is not expandable in a Taylor series around 0.

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = \begin{cases} e^{-\frac{1}{x^2}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

because derivatives of any order $\rightarrow 0$, but $f(x) \neq 0$
($f^{(n)}(0) = 0$, but $f(x) \neq 0$).

Application of Taylor series

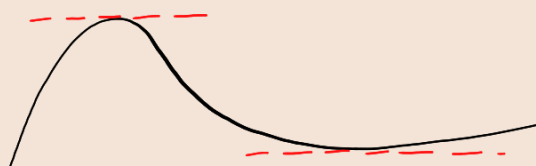
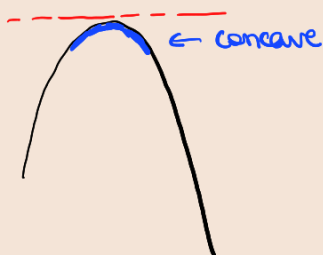
f diff. 2 times, $f'(x_0) = 0$

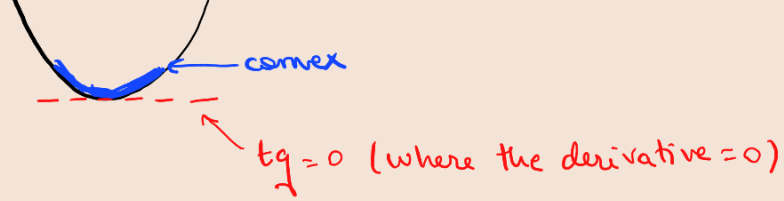
$$f(x) = f(x_0) + \frac{f''(c)}{2} \cdot (x - x_0)^2, \quad c \in (x, x_0)$$

$f''(x_0) > 0$ convexity $\Rightarrow x_0$ local min. $\cup$ $f'' > 0$ convex.

$f''(x_0) < 0$ concavity $\Rightarrow x_0$ local max. $\cap$ $f'' < 0$ concave

$\hookrightarrow$ without a table (faster way to tell if x_0 is $\begin{pmatrix} \min \\ \max \end{pmatrix}$).





Proof:

Taylor formula: $f(x) = T_1(x) + R_1(x)$

$$= \underbrace{f(x_0) + f'(x_0)(x-x_0)}_{T_1(x)} + \underbrace{\frac{f''(c)}{2}(x-x_0)^2}_{R_1(x)}, \quad c \in (x-x_0)$$

$$\Rightarrow f(x) = f(x_0) + \underbrace{\frac{f''(c)}{2}(x-x_0)^2}_{\geq 0}, \quad c \in (x, x_0)$$

$\Rightarrow x_0$ LOCAL MIN.

• if $f''(x_0)$ is > 0 , then $f(x) > f(x_0)$, for x around x_0

• if $f''(x_0)$ is < 0 , then $f(x) < f(x_0)$, for x around x_0
 $\Rightarrow x_0$ LOCAL MAX.

Theorem.

$$f'(x_0) = f''(x_0) = \dots = f^{(m-1)}(x_0) = 0, \quad f^{(m)}(x_0) \neq 0.$$

$\rightarrow m$ even, $f^{(m)}(x_0) > 0 \Rightarrow x_0$ local min. of f .

$\rightarrow m$ even, $f^{(m)}(x_0) < 0 \Rightarrow x_0$ local max. of f .

$\rightarrow m$ odd $\Rightarrow x_0$ not a local extremum point of f .

How to use libraries in Py for math purposes:

import matplotlib.pyplot as plt

import numpy as np

↓

domain of f : a vector

linspace() function generates nr. from an interval.

plot()

$x = (x_1, x_2, \dots, x_m)$

$y = (y_1, y_2, \dots, y_m)$

$\Rightarrow \dots$

$\$x\$ \rightarrow \text{Latex} \rightarrow \text{invented by Knuth.}$
 $[\text{Latex}]$

matrix $\rightarrow$ each row $\rightarrow$ a polynomial
 $\rightarrow$ each column $\rightarrow$?

$T[i,:]$ → access the whole row from the matrix

